

Table 9-6-A
BARRIER LENGTH OF NEED

Design Speed (mph)	Design Traffic Volume (ADT)				Shy Line Offset (ft)	Steepest Flare Rate for Barrier Inside Shy Line	Steepest Flare Rate for Barrier Beyond Shy Line	
	Over 10,000	5000-10,000	1000-5000	Under 1000			Rigid (Concrete Barrier)	Semi-Rigid or Flexible (W Beam, Thrie Beam)
	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)				
70	360	330	290	250	9.0	30:1	20:1	15:1
65	330	290	250	225	8.5	28:1	19:1	15:1
60	300	250	210	200	8.0	26:1	18:1	14:1
55	265	220	185	175	7.0	24:1	16:1	12:1
50	230	190	160	150	6.5	21:1	14:1	11:1
45	195	160	135	125	6.0	18:1	12:1	10:1
40	160	130	110	100	5.0	16:1	10:1	8:1
35	135	110	95	85	4.5	15:1	9:1	8:1
30	110	90	80	70	4.0	13:1	8:1	7:1

Equations to determine end of barrier need:

Flared Design

$$X = \frac{L_A + \frac{b}{a}(L_1) - L_2}{\frac{b}{a} + \frac{L_A}{L_R}}$$

$$Y = L_A - \frac{L_A}{L_R}(X)$$

Non-flared Design

$$X = \frac{L_R (L_A - L_2)}{L_A}$$

$$Y = L_2$$

Definition of Terms:

X = length of need in advance of the area of concern

Y = lateral offset from edge of traveled way

b/a = barrier flare (e.g., 1/30)

L_c = recommended clear zone (feet)

L_A = distance from edge of traveled way to back of hazard (feet)

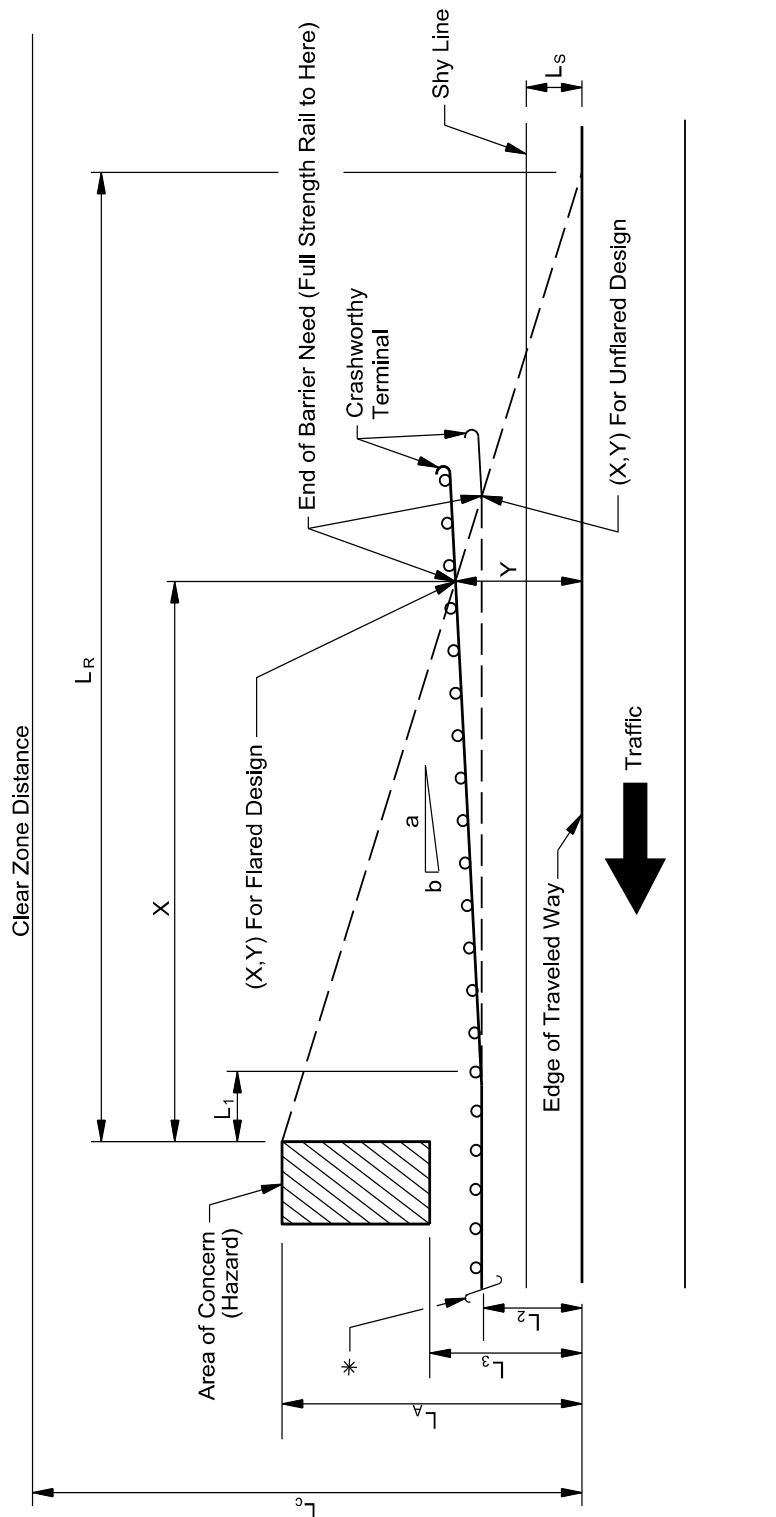
L_s = shy line offset, or distance at which barrier is no longer perceived as a hazard by a driver (feet)

L_R = runout length (feet)

L₁ = distance from hazard to where barrier flare begins (feet)

L₂ = distance from edge of traveled way to barrier (feet)

L₃ = distance from edge of traveled way to front of hazard (feet). (L₃ - L₂) should equal or exceed deflection distance.



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Figure 9-6-A

* The distance beyond the hazard should be determined by a length-of-need calculation for opposing traffic, if applicable.